Calculus II Week 1 Discussion Response

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0.1 Prompt

Create an integral for calculating the volume of a sphere (disk method). Use it to "prove" the formula for the volume of a sphere.

0.2 Response

Rewriting $(x-h)^2+(y-k)^2=r^2$, we have

$$(y-k)^2 = r^2-(x-h)^2 \implies y-k=\sqrt{r^2-(x-h)^2} \implies y=\sqrt{r^2-(x-h)^2}+k,$$

which gives the positive area of the circle. For the sake of calculating the volume of a sphere, we should assume constants $(h,k)=(0,0)$. Then,

$$y=\sqrt{r^2-x^2}$$

Then, the volume of one disk perpendicular to the x-axis for some x will have radius $r=\sqrt{r^2-x^2}$ and height dx. So, the volume of one disk is $\pi r^2 dx$. Then, the sum of all disks from $-r$ to r (and therefore the volume of a sphere with radius r) is

$$V=\pi\int_{-r}^{r} r^2 dx = \pi\int_{-r}^{r}\left[\sqrt{r^2-x^2}\right]^2 dx = \pi\int_{-r}^{r} r^2-x^2 dx$$

$$=\pi\left[r^2x-\frac{x^3}{3}\Big|_{-r}^{r}\right]=\pi\left[\left(r^2(r)-\frac{r^3}{3}\right)-\left(r^2(-r)-\frac{(-r)^3}{3}\right)\right]$$

$$=\pi\left[\left(\frac{3r^3-r^3}{3}\right)-\left(\frac{-3r^3-(-r^3)}{3}\right)\right]=\pi\left[\left(\frac{3r^3-r^3}{3}\right)-\left(\frac{-3r^3+r^3}{3}\right)\right]$$

$$=\pi\left[\left(\frac{2r^3}{3}\right)-\left(\frac{-2r^3}{3}\right)\right]=\boxed{\frac{4\pi r^3}{3}}$$